

SBT-DF203 Lab 4:

SMTP Email

Traffic Forensics

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(2025/FWSD/11479)

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Digital Forensics Investigation Report

Course Information

Item	Details
Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 4
Lab Title	SMTP Email Traffic Forensics
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026

1. Executive Summary

This investigation analysed SMTP email traffic contained in a packet capture file. The objective was to identify the email communication process, examine SMTP commands and server responses, recover authentication evidence, reconstruct the email transaction, and identify the systems involved.

The packet capture contained 60 packets associated with a single SMTP conversation. Analysis revealed successful authentication using the AUTH LOGIN mechanism, followed by the transmission of an email message from gurpartap@patriots.in to raj_deol2002in@yahoo.co.in. The SMTP session was established between the client host 10.10.1.4 and the mail server 74.53.140.153, which runs Exim SMTP services.

The investigation successfully identified the SMTP commands, authentication process, sender and recipient information, and key network metadata associated with the email transaction.

2. Authority, Scope and Evidence Description

2.1 Authority

This investigation was conducted as part of the requirements for SBT-DF203: Basic Networking Skills for Digital Forensics. All analysis was performed within an authorised Kali Linux virtual machine using the supplied SMTP packet capture.

2.2 Scope

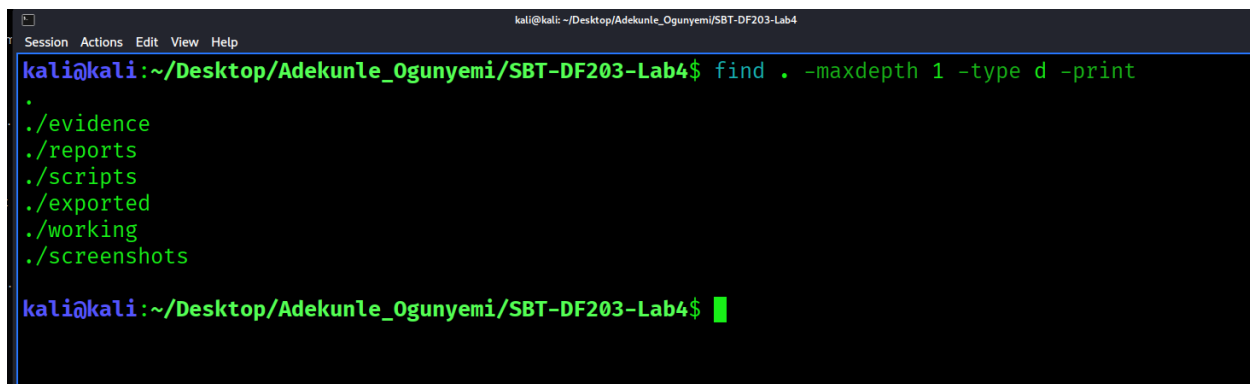
The scope of this investigation included:

1. SMTP traffic identification
2. SMTP command analysis
3. Authentication analysis
4. Base64 decoding

5. Email reconstruction
6. Network metadata extraction
7. Client and server identification
8. SMTP server assessment

2.3 Evidence Description

Item	Description
smtp.pcap	Original SMTP packet capture
smtp_working.pcap	Working copy used for analysis
Wireshark output	SMTP packet analysis
tshark output	Command and metadata extraction
Screenshots	Investigation evidence



```

kali@kali: ~/Desktop/Adekunle_Ogunyemi/GBT-DF203-Lab4
Session Actions Edit View Help
kali@kali:~/Desktop/Adekunle_Ogunyemi/GBT-DF203-Lab4$ find . -maxdepth 1 -type d -print
.
./evidence
./reports
./scripts
./exported
./working
./screenshots
kali@kali:~/Desktop/Adekunle_Ogunyemi/GBT-DF203-Lab4$

```

3. Environment and Acquisition Method

3.1 Working Environment

❖ Operating System:	Kali Linux
❖ Analysis Tool:	Wireshark
❖ Command-Line Tool:	tshark
❖ Scripting Tool: Python	3
❖ Hashing Tool:	sha256sum
❖ Evidence Type:	SMTP Packet Capture (PCAP)

3.2 Evidence Preparation

The supplied SMTP capture was preserved as the original evidence file. A working copy was created for analysis, and hashing procedures verified integrity before examination began.

```
(kali@kali)-[~/Desktop/Adekunle_Ogunyemi/GBT-DF203-Lab4]
$ cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
(kali@kali)-[~/Desktop/Adekunle_Ogunyemi/GBT-DF203-Lab4]
```

4. Evidence Integrity and Chain of Custody

4.1 Chain of Custody Worksheet

Field	Entry
Case Identifier	GBT-DF203-Lab4-AdekunleOgunyemi
Trainee Name	Adekunle Ogunyemi
ID Number	2025/FWSD/11479
Evidence File	smtp.pcap
Source	Training Packet Capture
Analysis System	Kali Linux
Notes	Working copy used for analysis

4.2 Evidence Integrity

The original evidence file was preserved and analysed using a separate working copy. This approach ensured that the original evidence remained unchanged throughout the investigation process.

5. Part A – SMTP Traffic Inventory

Analysis of the capture identified a total of 60 packets associated with the SMTP conversation. The SMTP traffic was reconstructed using TCP Stream 4.

```
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

EHLO GP

250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250-HELP

AUTH LOGIN

334 VXNlcm5hbWU6

Z3VycGFydGFwQHhhdHJpb3RzLmlu

334 UGFzc3dvcmQ0

cHVuamF1QDEyMw==

235 Authentication succeeded

MAIL FROM: <gurpartap@patriots.in>

250 OK

RCPT TO: <raj_deol2002in@yahoo.co.in>

250 Accepted

DATA

354 Enter message, ending with "." on a line by itself

From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="---=_NextPart_000_0004_01CA45B0.095693F0"
X-Mailer: Microsoft Office Outlook 12.0

.

19 client pkts, 10 server pkts, 18 turns.
```

The communication occurred between:

Item	Value
Client IP Address	10.10.1.4
Client Port	1470
Server IP Address	74.53.140.153
Server Port	25
TCP Stream Number	4
Packet Count	60

The SMTP server was identified through the initial service banner transmitted at the beginning of the session.

6. Part B – SMTP Command and Response Analysis

The SMTP session followed the normal sequence used during email transmission.

220-xc90.websitewelcome.com ESMTP Exim 4.69

This response indicated that the SMTP service was operational and ready to accept client connections.

SMTP Command Sequence

Command	Purpose
EHLO GP	Client identification
AUTH LOGIN	Authentication request
MAIL FROM	Sender address
RCPT TO	Recipient address
DATA	Begin email content
QUIT	Session termination

The SMTP server responded appropriately to each command and accepted the email transaction.

Important Response Codes

Code	Meaning
220	Service Ready
250	Request Accepted
334	Authentication Prompt
235	Authentication Succeeded
354	Begin Message Input

The successful completion of these exchanges confirms that the mail server accepted the email.

7. Part C – Authentication Analysis

The SMTP session used the AUTH LOGIN authentication method.

The username and password were transmitted as Base64-encoded values and were successfully decoded using an offline Python script.

Decoded Credentials

Item	Value
Username	gurpartap@patriots.in
Password	punjab@123

Redacted Credentials

Item	Value
Username	g*****@patriots.in
Password	p*****3

The server returned the following response after authentication:

235 Authentication succeeded

This confirmed that the supplied credentials were accepted by the mail server.

Base64 encoding provided no security protection because the values could be easily reversed and recovered.

8. Part D – Email Reconstruction

The SMTP transaction identified both the sender and recipient addresses.

Envelope Sender

MAIL FROM: gurpartap@patriots.in

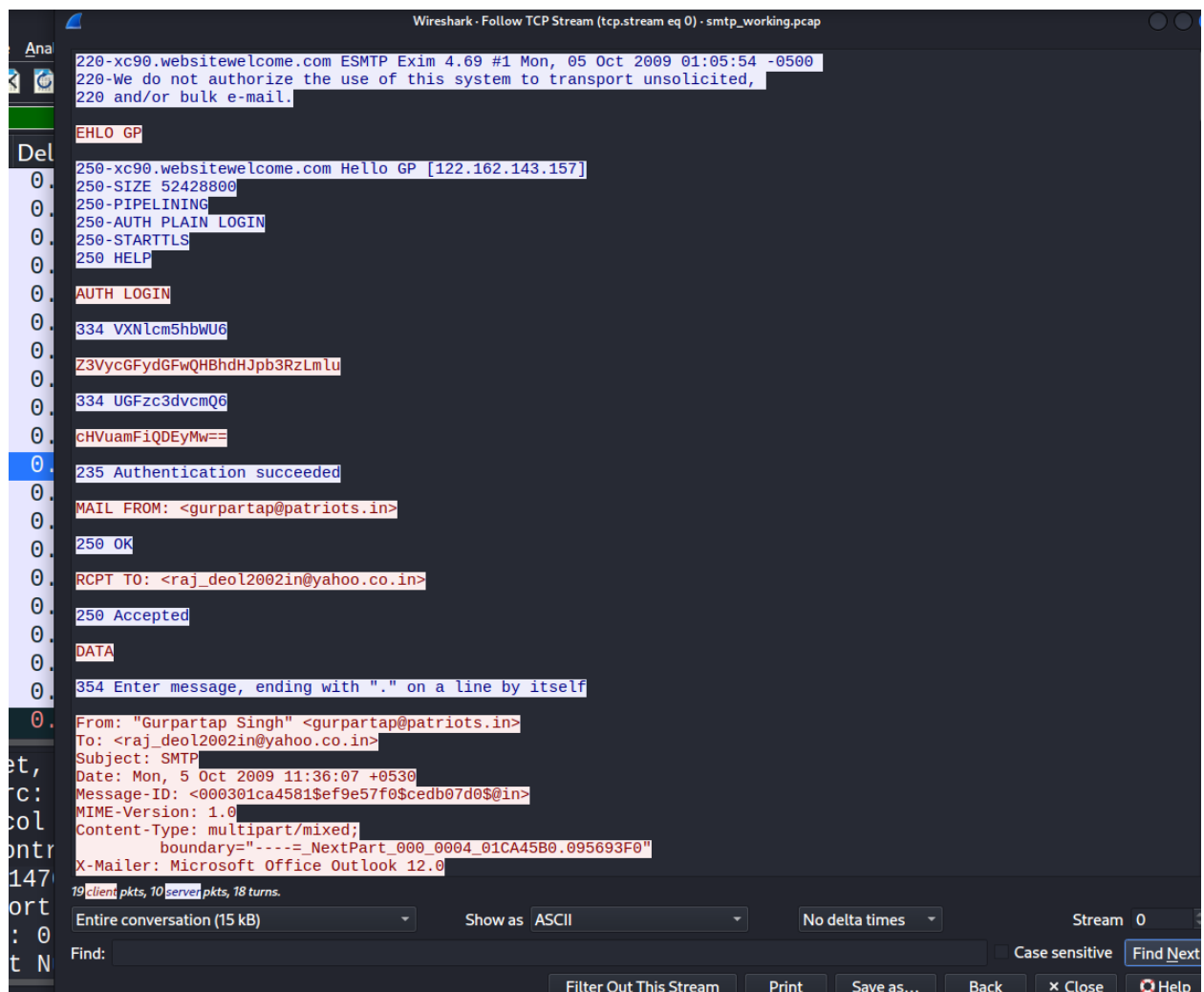
Envelope Recipient

RCPT TO: <raj_deol2002in@yahoo.co.in>

The server accepted both addresses and allowed the email body to be transmitted after the DATA command.

354 Enter message, ending with "." on a line by itself

The capture confirmed successful message submission from the sender to the recipient.



```
Wireshark - Follow TCP Stream (tcp.stream eq 0) - smtp_working.pcap

220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

EHLO GP

250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

AUTH LOGIN

334 VXNlcm5hbWU6

Z3VycGFydGFwQHhhdHJpb3RzLmlu

334 UGFzc3dvcmQ6

cHVuamF1QDEyMw==

235 Authentication succeeded

MAIL FROM: <gurpartap@patriots.in>

250 OK

RCPT TO: <raj_deol2002in@yahoo.co.in>

250 Accepted

DATA

354 Enter message, ending with "." on a line by itself

From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="====_NextPart_000_0004_01CA45B0_095693F0"
X-Mailer: Microsoft Office Outlook 12.0

19 client pkts, 10 server pkts, 18 turns.

Entire conversation (15 kB) Show as ASCII No delta times Stream 0
Find: Case sensitive Find Next
Filter Out This Stream Print Save as... Back x Close Help
```

9. Part E – Client, Server and Network Metadata

The SMTP communication involved the following systems.

Item	Value
SMTP Server Hostname	xc90.websitewelcome.com
Mail Server Software	Exim 4.69
Server IP Address	74.53.140.153
Server Port	25
Client IP Address	10.10.1.4
Client Port	1470
Authentication Method	AUTH LOGIN
TCP Stream Number	4
Packet Count	60

The client established a connection to the SMTP server and completed authentication before transmitting the email.

10. Part F – Encryption Assessment and Limitations

The SMTP commands, authentication exchange, and email delivery information were visible within the packet capture.

The presence of readable AUTH LOGIN exchanges and recoverable credentials indicates that the communication was not protected by TLS encryption at the stage captured in the packet trace.

If STARTTLS or other encryption mechanisms had been used, the email content and authentication data would not have been visible in plaintext.

11. Required Forensic Findings

Finding	Observation
SMTP Session	Observed
Packet Count	60
TCP Stream Number	4
Client IP Address	10.10.1.4
Client Port	1470
Server IP Address	74.53.140.153
Server Port	25
SMTP Server Hostname	xc90.websitewelcome.com
Mail Server Software	Exim 4.69
SMTP Banner	220-xc90.websitewelcome.com ESMTP Exim 4.69
Authentication Method	AUTH LOGIN
Authentication Result	235 Authentication Succeeded
Sender Address	gurpartap@patriots.in
Recipient Address	raj_deol2002in@yahoo.co.in
DATA Command Observed	Yes
Message Transmission	Successful
STARTTLS Observed	Not observed in available evidence
Key Limitation	Full email headers and message body not available in supplied evidence

12. Limitations and Challenges

Challenge 1 – Sensitive Authentication Data

The packet capture contained recoverable authentication credentials. These values were treated as sensitive evidence and were masked within the report.

Challenge 2 – Limited Message Content

Although the SMTP commands and addressing information were visible, the supplied evidence did not include the full reconstructed message headers or message body.

General Limitations

- ❖ Only information contained within the packet capture can be analysed.
- ❖ Missing packets may affect complete message reconstruction.
- ❖ Encryption would prevent recovery of message contents.
- ❖ Authentication credentials recovered during analysis must be handled as confidential evidence.

13. Conclusion and Recommendations

13.1 Summary of Findings

- ❖ SMTP traffic was successfully identified and analysed.
- ❖ The SMTP server was identified as xc90.websiteswelcome.com running Exim 4.69.
- ❖ Client authentication was performed using AUTH LOGIN.
- ❖ Authentication succeeded using the supplied credentials.
- ❖ The email originated from gurpartap@patriots.in.
- ❖ The email was addressed to raj_deol2002in@yahoo.co.in.
- ❖ Network metadata was successfully recovered.
- ❖ SMTP commands and server response codes were documented.

13.2 Recommendations

- ❖ Implement TLS encryption to protect SMTP authentication exchanges.
- ❖ Avoid transmitting credentials in a form that can be easily decoded.
- ❖ Preserve original packet captures before analysis.
- ❖ Maintain proper evidence handling procedures throughout the investigation.

- ❖ Document all limitations and assumptions within the final forensic report.

14. Academic Integrity Statement

I confirm that I completed this lab in an authorised environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.